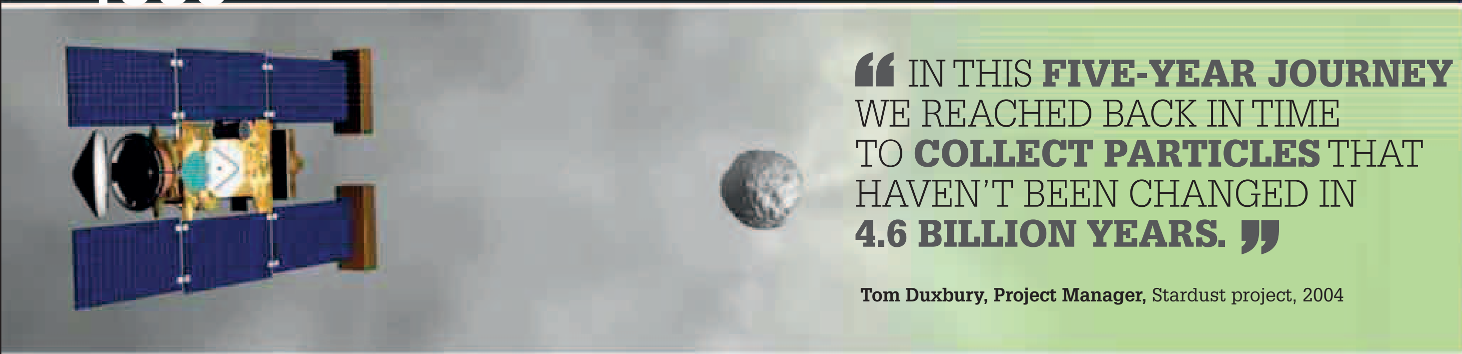
1999



> IN THIS **FIVE-YEAR JOURNEY** WE REACHED BACK IN TIME TO **COLLECT PARTICLES** THAT HAVEN'T BEEN CHANGED IN **4.6 BILLION YEARS.**

Tom Duxbury, Project Manager, Stardust project, 2004

NASA's Stardust probe flew past comet Wild 2 and a retractable aerogel panel collected the dust from the comet's coma (dust cloud).

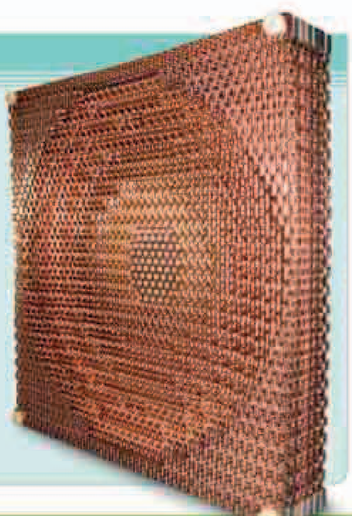
IN FEBRUARY, NASA's STARDUST PROBE BEGAN an unprecedented mission: to **collect samples** from the dust cloud (coma) surrounding a **comet's nucleus**. As it traveled, it also collected interstellar dust. When Stardust flew past Earth in 2006, it released a sample-return capsule, which returned to Earth.

Scientists **created three identical goats** by transferring the contents of embryo cells into empty egg cells. The embryonic cells **had an extra gene**, which caused the goats to produce a blood clotting factor that could be harvested from their milk and **used in human medicines**.

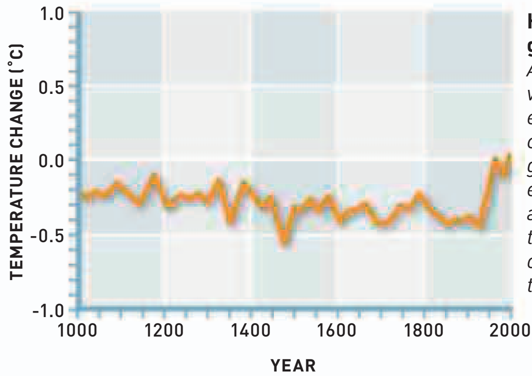
The increasing number of digital gadgets around the home encouraged the **development of wireless data connections**. The most widely used wireless networking protocol—IEEE 802.11 (see 1997)—was given a user-friendly name this year: Wi-Fi. For one-to-one connections between various types of device, a **new protocol was released**: Bluetooth. Developed by Swedish company Ericsson in 1994, the most popular use for Bluetooth is sending digital audio signals to wireless headphones or speakers.

3 THE NUMBER OF **IDENTICAL GOATS** PRODUCED BY **CLONING** AN **ADULT** ANIMAL

METAMATERIALS



Metamaterials are man-made materials that have properties not found in naturally occurring materials. One example (right) is this material with tiny metal coils embedded within it, which can divert microwaves around itself. This makes the material invisible to the microwaves. A metamaterial that does the same to visible light could act as an invisibility cloak.



Hockey-stick graph
A simplified version (without error bars) of the controversial graph that shows estimated average global temperature over the past thousand years.

American climatologists Michael Mann (b.1965) and Raymond Bradley (b.1948) and American dendrochronologist (studies tree rings) Malcolm Hughes (b.1943) constructed a graph of the **estimated average global temperature** over the past thousand years. The data were derived from meteorological readings, with older estimates based on historical records and tree rings. The graphs showed a gradual cooling, with a **rapid rise in temperature** corresponding to the rise in global population and industrialization. The **graph's shape** reminded American climate scientist Jerry Mahlman (1940–2012) of a **hockey stick**. To climate scientists, the graph is a symbol of the **effect of human activity** on the climate—and a stark reminder of the need to reduce carbon dioxide emissions. However, many who were opposed to the idea that human activity is causing global warming disputed the accuracy of the graph.

Researchers studying growth hormones **discovered a protein**, which they called ghrelin, that is secreted by cells in the stomach lining. Ghrelin acts on the **appetite center in the brain** (see leptin, 1994). The release of ghrelin into the bloodstream is determined by stretch receptors in the stomach lining: when the stomach is full, less ghrelin is produced, but as the stomach empties between meals, more ghrelin is produced, **leading to the sensation of hunger**.

Also this year, British physicist **John Pendry** (b.1943) described a new class of materials with **unusual properties**, called metamaterials. One metamaterial in particular held the promise of invisibility (see panel, left).

2000

IN THE MONTHS LEADING UP TO THE TURN OF THE MILLENNIUM, people were warned of the possibility that **electronic devices and systems could fail**, as computer operating systems used two digits to represent years and internal clocks would revert to the year 1900. In a world in which people's livelihoods and even their safety increasingly rely upon computerized systems, this possibility—known as the **Y2K problem or the Millennium Bug**—led to widespread panic. Software engineers worked hard to ensure that the fears would turn out to be unfounded, and indeed only a small number of systems were affected.

Fruit fly
The fruit fly has been used as a model organism in genetic research since 1910, and so was a natural choice to have its genome sequenced.

January 22 Italian cell biologist Angelo Vescovi announces the creation of **blood cells from brain stem cells** in mice

February 7 Launch of NASA's Stardust mission to **asteroid 5535 Annefrank and comet Wild 2**

April 23 Publication of the controversial graph showing the recent **rise in the average global temperature**

April 27 Scientists **produce three clones** of an adult goat

July 31 NASA's Lunar Prospector is **deliberately crashed into the lunar north pole**, revealing water ice in the collision debris

November Physicists publish details of the first metamaterials, opening up the **possibility of invisibility cloaking**

December 1 Ericsson releases the **wireless networking protocol Bluetooth**, which they invented in 1994

December 9 Appetite-regulating hormone ghrelin discovered

January 14 Creation of vitamin A-rich **transgenic rice** announced